UAV component/feature	Function in agriculture	Benefit to farmers
	documentation	abnormalities
Multispectral sensors (e.g., NDVI)	Plant health, vigor, and stress monitoring	Data-driven crop management; targeted interventions
Thermal cameras	Irrigation monitoring and moisture stress detection	Prevents water waste and improves drought resilience
AI/ML analytics platforms	Automated disease, pest, and nutrient diagnosis	Reduces expert dependency and time delay in analysis
Precision spraying drones	Targeted pesticide, fertilizer, and micronutrient application	Lower chemical usage, reduced cost, and improved environmental safety
Autonomous flight planning systems	Automated path planning and repeatable surveys	Consistent monitoring and reduced labor requirements

8.4.1 Precision farming and yield enhancement

Precision farming focuses on optimizing agricultural efficiency by tailoring crop management to the specific conditions of each field section. UAVs equipped with high-resolution imaging systems, such as RGB, multispectral, hyperspectral, and thermal cameras, enable farmers to observe spatial variations in plant health that are not visible from the ground. This data helps detect nutrient deficiencies, pest infestations, disease spread, and growth anomalies at early stages, allowing timely corrective interventions.

In yield optimization, drones support variable-rate application strategies, where fertilizers, seeds, and irrigation are distributed only where needed, reducing waste and improving overall productivity. By continuously